
General Chemistry**Lec(2): Structure of the Atom****A historical overview of the development of the concept of the atom:**

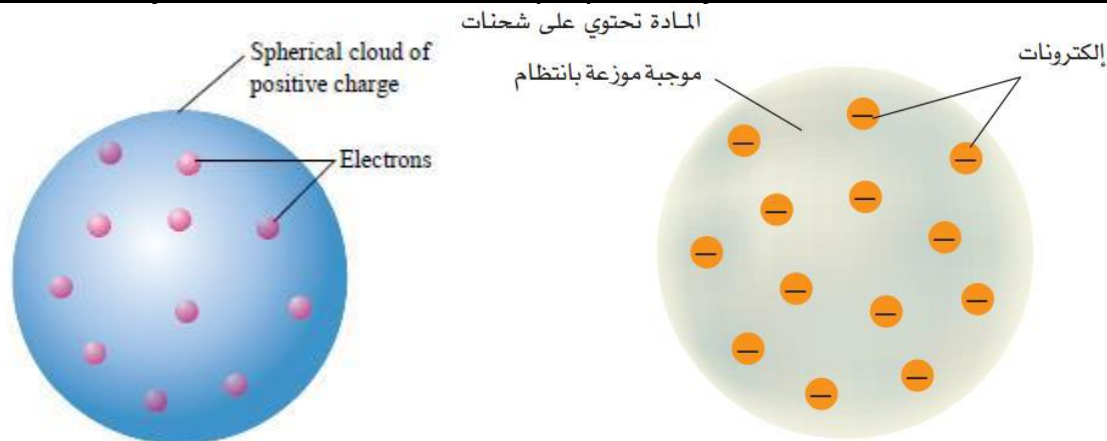
The concept of the atom developed gradually according to the beliefs of philosophers. It began with the Babylonians, who considered water to be the basis of life. Then came Aristo, who proposed that matter is composed of water, air, earth, and fire, until Democritus proposed that matter is not divisible infinitely and suggested that matter is composed of small parts called atoms. The word atom comes from the Greek word **Atomos**, meaning that which cannot be divided, and this concept continued until the beginning of the nineteenth century, when the true nature of matter as having an atomic nature appeared, when the English scientist John Dalton announced his atomic theory in 1808, which represents the following postulates:

- Matter is composed of extremely small particles called atoms, which are indivisible.
- An element is a substance that cannot be divided into other substances.
- Atoms of the same element are identical in their properties such as shape and size, and differ from atoms of other elements.
- Atoms of one element cannot be transformed into atoms of another element through chemical reactions.
- In chemical reactions, atoms combine with one another, separate from one another, or change their positions to form molecules.

Then this theory was followed by models to explain the structure of the atom, among which are the following:

Thomson's model of atomic structure:

In 1897, the scientist Thomson proposed that the atom is a sphere of positive electric charge penetrated by a number of electrons equal to the total positive charge, and the atom is electrically neutral because the number of electrons equals the number of positive charges. This model was called as plum pudding model. The electrons present the plums.



This theory explained some phenomena such as the photoelectric effect, the kinetic theory of gases, and the emission of electromagnetic waves, but it failed to explain the details of alpha rays and emission spectra.

Limitation: could not account for all properties of atoms. Rutherford scattering experiment raised an objection against the model. Merits : it accounted for the electric neutrality of atoms.

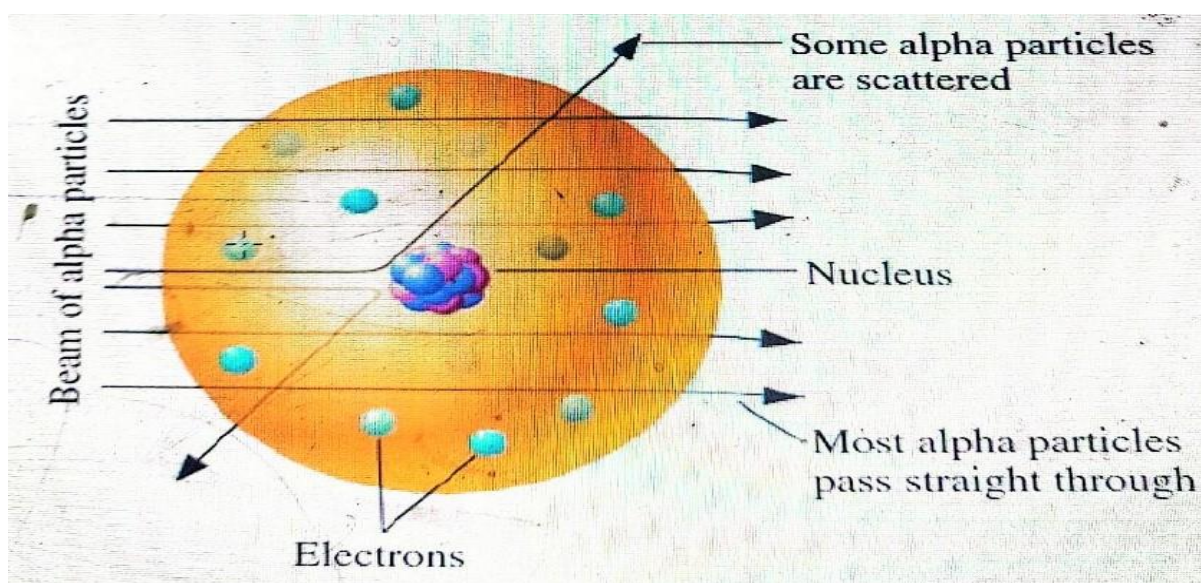
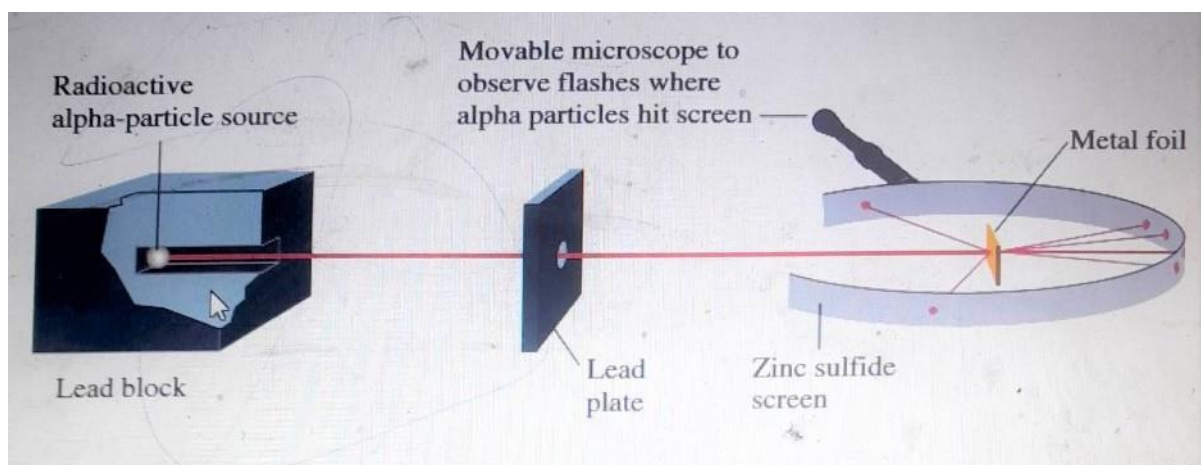
Rutherford's model:

The English scientist Rutherford proposed that the atom consists of a material nucleus carrying a positive charge, around which electrons revolve. The scientist Moseley then completed this picture when he found that the number of electrons revolving around the nucleus is equal to the number of protons inside the nucleus, so that the atom is electrically neutral. To prove his conception, in 1911 he conducted an experiment on a thin sheet of gold by exposing it to alpha particles to measure the amount of their deflection. The results showed that about 99% of alpha particles penetrated the sheet without changing their path, while a small number were deflected at different angles, and a very small number were reflected and did not penetrate the sheet.

From this, Rutherford concluded:

- Most of the volume of the atom is empty space, which explains the passing of most alpha particles (99%) without deflection.
- The mass of the atom is concentrated in a very small region at the center of the atom, which he called the nucleus, and this explains the reflection of a small number of alpha particles when they approach this very narrow region.
- The nucleus of the atom carries a positive charge, as evidenced by the deflection of some alpha particles at different angles.

- Since the atom is electrically neutral, the number of electrons surrounding the nucleus must equal the amount of positive charge.
- proposed the existence of neutral particles inside the nucleus of the atom, which were later identified In 1932, by English scientist James Chadwick and called neutrons.



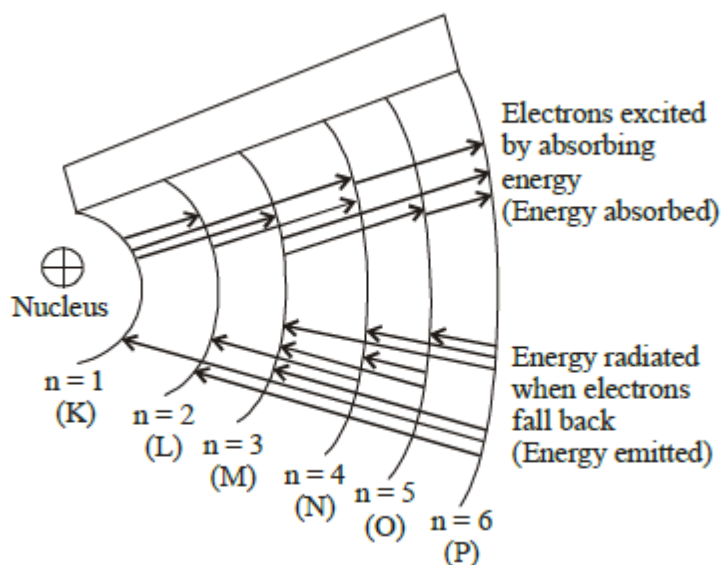
Drawbacks of Rutherford's model atom.

- Rutherford model of atom could not explain the stability of the atom.
- Rutherford model of atom could not explain as to how the electrons are distributed in the extra nuclear portion in an atom

Bohr's model:

In 1913, the Danish physicist Niels Bohr proposed a model for the structure of the atom, more developed than Rutherford's model, and to explain the hydrogen spectrum. The important postulates on which Bohr's model is based are following:

- The electron revolves in circular orbits called energy levels, designated by the numbers 1, 2, 3, 4... which are called principal quantum numbers and symbolized by K, L, M, N, O, P, Q.
- The electron does not emit any energy as long as it moves in its specified orbit at a given energy level.
- The atom does not radiate energy, but when it absorbs a certain amount of energy, the electron moves from a lower-energy orbit to a higher-energy orbit, or emits radiation when it returns, and this change in energy equals the difference between the energies of the two orbits.

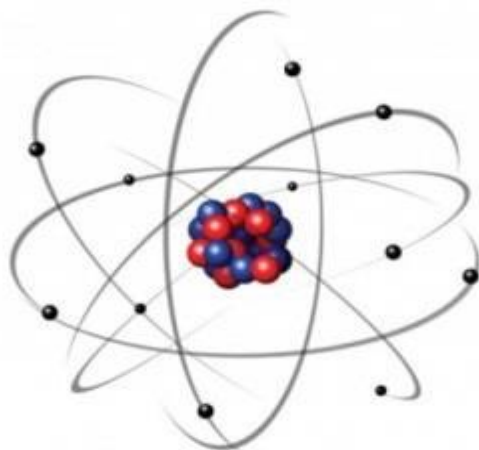


The emission or absorption of energy in the form of radiation can only occur when an electron jumps from one stationary orbit to another.

$$\Delta E = E_{\text{final}} - E_{\text{initial}} = h\nu$$

Where h is Planks constant $h = 6.625 \times 10^{-34} \text{ Js}$, ν is frequency of emitted or absorbed radiation.

Energy is absorbed when the electron moves from lower to higher orbit and is emitted when it jump from higher to lower orbit

**Failures/Limitations of Bohr's theory:**

- Bohr's theory cannot explain the spectra of elements more complex than hydrogen.
- Bohr assumed that the position and velocity of the electron could be determined at the same time, which was later shown to be scientifically impossible.
- He assumed that the electron moves in a flat circular path around the nucleus, whereas it was later discovered that the atom exists in three-dimensional space.
- He assumed that the electron is a negative particle and did not take into account its wave properties.

Modern atomic theory:

- In 1924, the French scientist Louis de Broglie proposed that the electron has a wave nature.
- The German physicist Werner Heisenberg introduced the uncertainty principle, which states that it is impossible to determine both the position and momentum (or velocity and energy) of the electron at the same time with precision.
- In 1926, the Austrian physicist Erwin Schrödinger developed a wave equation that describes the wave motion of the electron in the atom and its energy in general.

Components of the atom:

The atom is very small in size; although its diameter is approximately 10^{-10} meters, the diameter of its nucleus is about 10^{-15} meters. The atom consists of a number of fundamental and elementary particles known as elementary particles, whose number has reached hundreds. These particles have been classified into groups according to their mass, nature, type of interaction, and symmetry, such as leptons and hadrons. We explain some of them as follows:

1. Hadrons:

They include particles that are stable and others that are unstable, and they comprise two groups: mesons and baryons.

a. Baryons:

The name comes from the Greek word *bary*, meaning heavy. They are strongly interacting particles composed of three quarks. Some are stable, such as protons and neutrons, and some are unstable.

- **Proton (p^+):** First observed by Goldstein in 1886. Rutherford later named it the proton. Its charge is equal in magnitude to that of the electron but opposite in sign, and its mass is approximately 1.673×10^{-27} kg.
- **Neutron (n):** A neutral, electrically uncharged particle discovered by the English scientist James Chadwick in 1932. Its mass is approximately 1.675×10^{-27} kg. There is also a particle called the deuteron, which consists of a proton and a neutron.

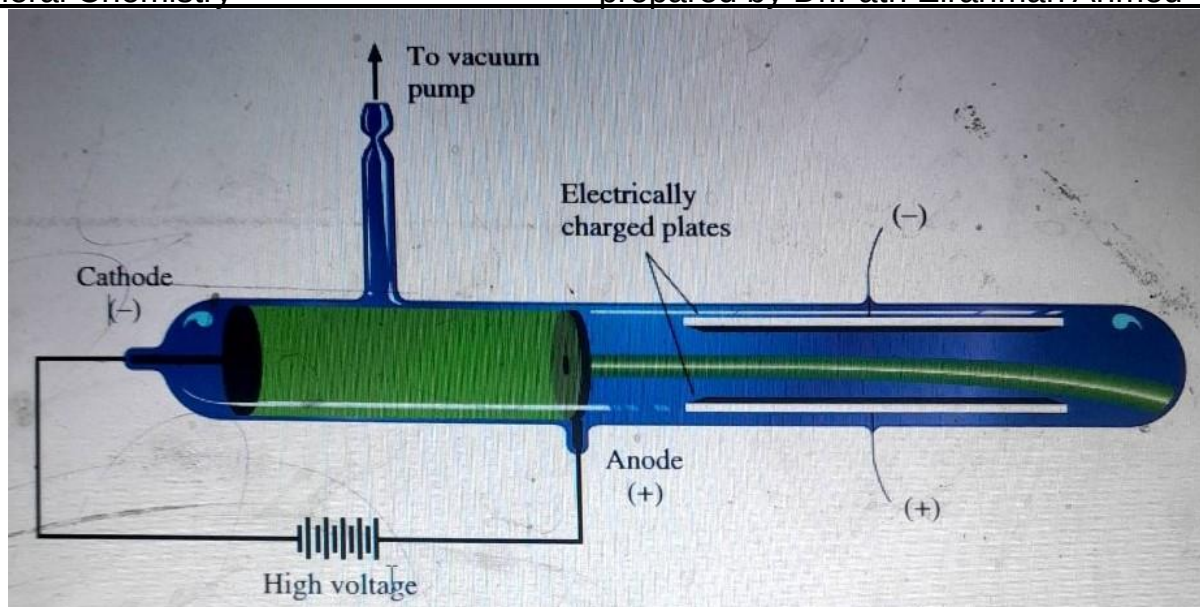
b. Mesons:

They are unstable particles whose masses range between that of the electron and the proton. They are composed of a quark and an antiquark. Examples include pion (π) mesons and muon (μ) mesons. The pion has a mass about 300 times that of the electron and exists in three charge states.

Leptons:

They are particles considered the basic components of matter. There are six leptons, the most famous of which are electrons.

- **Electron:** The English scientist G. Stoney first proposed the existence of small electric particles in 1891 and named them electrons. The electron is the smallest particle carrying a negative charge. In 1897, J. J. Thomson determined the charge-to-mass ratio (e/m) of the electron by measuring the deflection of cathode rays in electric and magnetic fields. In 1909, Millikan measured the charge of the electron as 1.6022×10^{-19} coulombs. From the ratio e/m , the mass of the electron can be calculated as 9.109×10^{-31} kg.



Fundamental Particle	Mass	Charge	Relative Charge
Electron	$9.11 \times 10^{-31} \text{ Kg}$	$-1.602 \times 10^{-19} \text{ Coulomb}$	-1
Proton	$1.672 \times 10^{-27} \text{ Kg}$	$+1.602 \times 10^{-19} \text{ Coulomb}$	+1
Neutron	$1.675 \times 10^{-27} \text{ Kg}$	0	0

Properties subatomic particles

The sub-atomic particles ; electrons, protons and neutrons are called fundamental particles of all matters.

- Electron e : Electron is a fundamental sub-atomic particle having negligible mass of $9.11 \times 10^{-31} \text{ kg}$ and carrying a charge of $-1.602 \times 10^{-19} \text{ Coulomb}$.
- Proton p: Proton is a fundamental sub-atomic particle having mass of $1.672 \times 10^{-27} \text{ kg}$ and carrying a charge of $+1.602 \times 10^{-19} \text{ Coulomb}$.
- Neutron n: Neutron is a fundamental sub-atomic particle having mass of $1.675 \times 10^{-27} \text{ Kg}$ and carrying no charge

Atomic number (Z):

It is the number of protons present in the nucleus of an atom of an element.

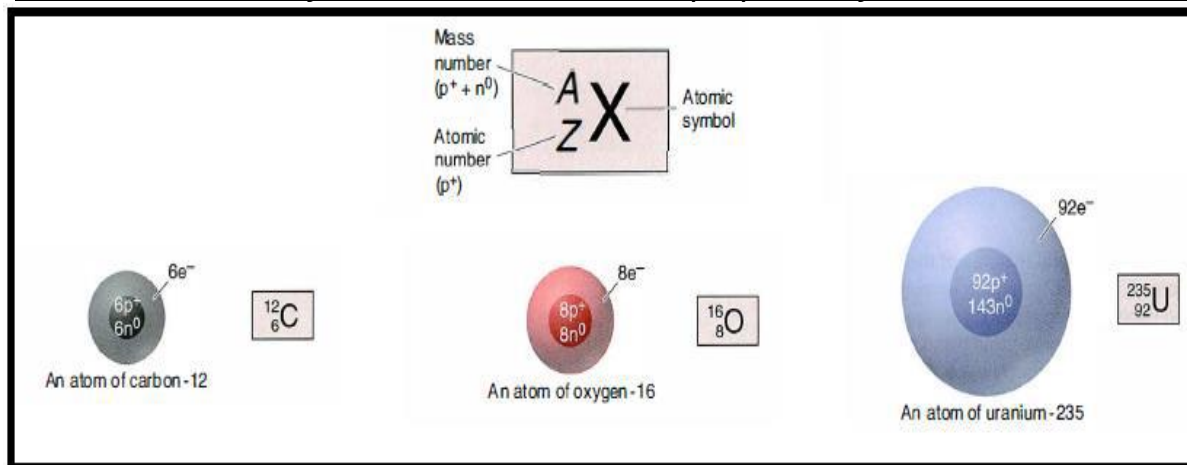
Mass number (A):

It is the total number of protons and neutrons and was formerly called the atomic weight.

$A = \text{number of protons (Z)} + \text{number of neutrons}$

$\text{Number of neutrons} = A - Z$

The mass number and atomic number are written with the symbol of the element as follows:



Calculation of number of electron:

For neutral atom Number of Electrons = Z

For positive ions (cation). Number of Electrons = $Z - \text{charge on cation}$

For negative ions(anion) Number of Electrons = $Z + \text{charge on Anion}$

Example:

1) Calculate the number of protons, neutrons and electrons in $^{80}_{35}\text{Br}$?

Solution

Here, $Z = 35$, $A = 80$

No. of protons = Atomic Number = 35

No. of neutrons = $A - Z = 80 - 35 = 45$

As the atom is neutral, No. of electrons = No. of protons = 35.

2) Calculate the number of electron, protons and neutrons, in $^{31}_{15}\text{P}^{-1}$

Solution:

Here $Z = 15$ $A = 31$

Number of protons = $Z = 15$

Number of neutrons = $A - Z = 31 - 15 = 16$

Number of electrons for anions = $Z + \text{charge on anion} = 15 + 3 = 18$

Isotopes:

They are atoms of the same element that have the same atomic number but different mass numbers, meaning they have the same number of protons but differ in the number of neutrons. This similarity in the number of protons makes them similar in chemical properties. They are also called nuclides.

Isotopes			Elements
${}^3_1\text{H}$	${}^2_1\text{H}$	${}^1_1\text{H}$	Hydrogen
${}^1_6\text{C}$	${}^1_6\text{C}$	${}^{12}_6\text{C}$	Carbon
${}^{18}_8\text{O}$	${}^{17}_8\text{O}$	${}^{16}_8\text{O}$	Oxygen
${}^{24}_{11}\text{Na}$	${}^{23}_{11}\text{Na}$	${}^{22}_{11}\text{Na}$	Sodium
${}^{35}_{17}\text{Cl}$	${}^{37}_{17}\text{Cl}$	${}^{38}_{17}\text{Cl}$	Chlorine
${}^{238}_{92}\text{U}$	${}^{235}_{92}\text{U}$	${}^{234}_{92}\text{U}$	Uranium

Properties of Isotopes: They are

- ☐ atoms of the same element.
- ☐ have the same atomic no. but different mass number.
- ☐ have the same no. of protons and electrons.
- ☐ differ in their no. of neutrons.
- ☐ have different physical properties such as m.pt., b.pt, density, viscosity, etc.

Relative atomic mass(RAM): the atomic mass is often contain a fraction which stem from taking the different atomic masses of isotopes of element. This give us the relative mass which is given by the following formula:

$$\text{RAM} = \frac{(A_1 \times n_{a1}) + (A_2 \times n_{a2}) + (A_3 \times n_{a3}) + \dots \text{etc}}{100}$$

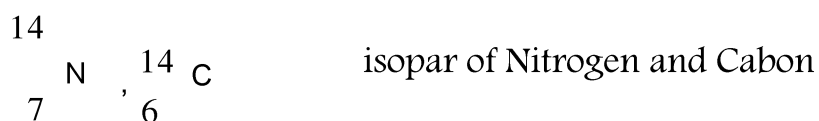
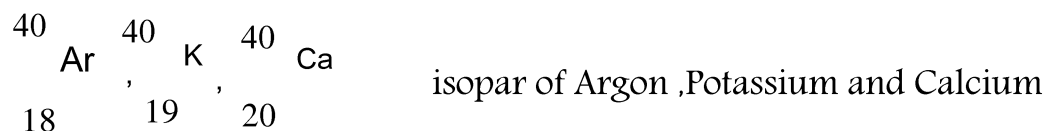
Where : $A_1, A_2, A_3 \dots \text{etc}$ are the atomic masses of isotopes 1, 2, 3...etc of the element $n_{a1}, n_{a2}, n_{a3} \dots \text{etc}$ are the natural percentage abundance of each isotope.

Example: Sodium (Na) has two isotope ${}^{22}\text{Na}$ and ${}^{23}\text{Na}$ with natural percentage abundance of 2.9% and 97.1% . calculate the RAM of boron.

$$\text{RAM} = \frac{(22 \times 2.9) + (23 \times 97.1)}{100} = 22.97$$

Isobars:

They are atoms of different elements that differ in atomic number but have the same mass number, such as calcium, argon, carbon, and nitrogen.

**Isotones:**

They are atoms of different elements that have the same number of neutrons but different atomic and mass numbers.

no. of neutron	Atomic number (no. of proton)	Atomic mass	Element
16	14	30	${}^{30}_{14}\text{Si}$
16	15	31	${}^{31}_{15}\text{P}$
16	16	32	${}^{32}_{16}\text{S}$

The atom is electrically neutral because the number of protons in the nucleus equals the number of electrons revolving around it.